

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Scala For Data Science

Practical X

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Class: SYBSc IT	Subject: Scala For Data Science
Date of Assignment: 4/08/2026	Date/Time of Submission: 4/08/2026

AIM: Implement linear regression using Breeze. Fit a model to a small dataset and predict a value.

CODE:

```
import breeze.linalg._
import breeze.stats._
import breeze.nummerics._

object Demo extends App{
  val x = DenseVector(1.0,2.0,3.0,4.0,5.0)
  val y = DenseVector(2.0,4.1,6.0,8.1,9.9)
  val mX = mean(x)
  val mY = mean(y)
  val n = sum((x-mX) ** (y-mY))
  val d = sum((x-mX) *(x-mX))
  val s = n/d
  val i = mY - s * mX
  println(f"Slope: $s%.2f")
  println(f"Intercept: $i%.2f")
  val nx = 6.0
  val p = s * nx + i
  println(f"Predicted value = $p%.2f")
  println("IQRA S021")
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe
Slope: 1.98
Intercept: 0.08
Predicted value = 11.96
IQRA S021

Process finished with exit code 0
```

Practical XI

AIM: Calculate mean, median, and mode of a list of numbers. Implement basic statistical calculations using Scala collections.

CODE:

```
build.sbt  PRACTICALX.scala x
1  object Demo extends App{
2      val numbers = List(2,4,4,6,8,4,10)
3      val m = numbers.sum.toDouble / numbers.length
4      val s = numbers.sorted
5      val median = s(s.length / 2)
6      val d = numbers.distinct
7      var mode = d(0)
8      var maxcount = 0
9      for (x <- d){
10         val count = numbers.count(_ == x)
11         if (count > maxcount){
12             maxcount = count
13             mode = x
14         }
15     }
16     println("Numbers: "+numbers)
17     println("Mean: "+m)
18     println("Median: "+median)
19     println("Mode: "+mode)
20     println("IQRA S021")
Demo
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Numbers: List(2, 4, 4, 6, 8, 4, 10)
Mean: 5.428571428571429
Median: 4
Mode: 4
IQRA S021

Process finished with exit code 0
```

Practical XII

AIM: Write a program to compute frequency distribution and cumulative frequency of a dataset.

CODE:

```
object Demo extends App{

    val data = List(3,5,3,2,5,4,3,4,5)

    val freq = data.distinct.map(x => (x,data.count(_ ==x)))
    val sor = freq.toSeq.sortBy(_._1)
    var cum = 0
    println("Value\tFreq\tCumulative")

    for((value,count) <- sor) {
        cum += count
        print(s"$value\t\t$count\t\t$cum")
    }
    println("\nIQRA S021")
}
```

OUTPUT:

```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
Value   Freq   Cumulative
2       1      13      3      44      2      65      3      9
IQRA S021

Process finished with exit code 0
```

Practical XIII

AIM: Filter rows in a dataset where a specific column value exceeds a threshold.

CODE:

```
1  import scala.io.Source
2  ▶ object Demo extends App{
3      val file = Source.fromResource("iris.csv")
4      println("Sepal Length Greater than 5.0:")
5      for(line <- file.getLines().drop(1)) {
6          val row = line.split(",")
7          val sepalLength = row(0).toDouble
8          if (sepalLength > 6.0) {
9              println(line)
10         }
11     }
12     file.close()
13     println("\nIQRA S021")
14 }
15
```

OUTPUT:

1	sepal_length,sepal_width,petal_length,petal_width,species
2	5.1,3.5,1.4,0.2,~setosa
3	4.9,3.0,1.4,0.2,~setosa
4	4.7,3.2,1.3,0.2,~setosa
5	4.6,3.1,1.5,0.2,~setosa
6	5.0,3.6,1.4,0.2,~setosa
7	5.4,3.9,1.7,0.4,~setosa
8	4.6,3.4,1.4,0.3,~setosa
9	5.0,3.4,1.5,0.2,~setosa
10	4.4,2.9,1.4,0.2,~setosa
11	4.9,3.1,1.5,0.1,~setosa
12	5.4,3.7,1.5,0.2,~setosa
13	4.8,3.4,1.6,0.2,~setosa
14	4.8,3.0,1.4,0.1,~setosa
15	4.3,3.0,1.1,0.1,~setosa
16	5.8,4.0,1.2,0.2,~setosa
17	5.7,4.4,1.5,0.4,~setosa
18	5.4,3.9,1.3,0.4,~setosa
19	5.1,3.5,1.4,0.3,~setosa
20	5.7,3.8,1.7,0.3,~setosa
21	5.1,3.8,1.5,0.3,~setosa
22	5.4,3.4,1.7,0.2,~setosa
23	5.1,3.7,1.5,0.4,~setosa